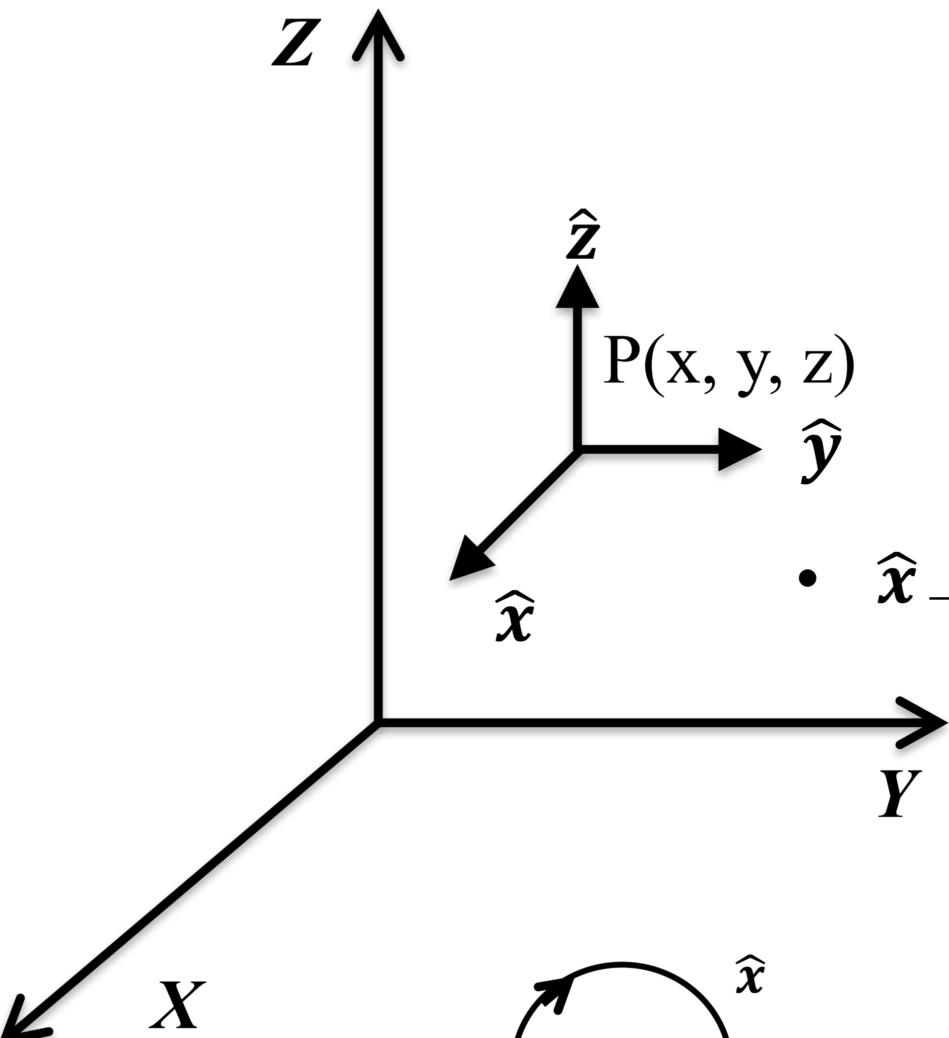
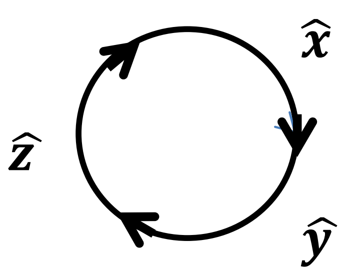
Lecture 04

Coordinate systems

Continued…

Cartesian co-ordinate system in 3D

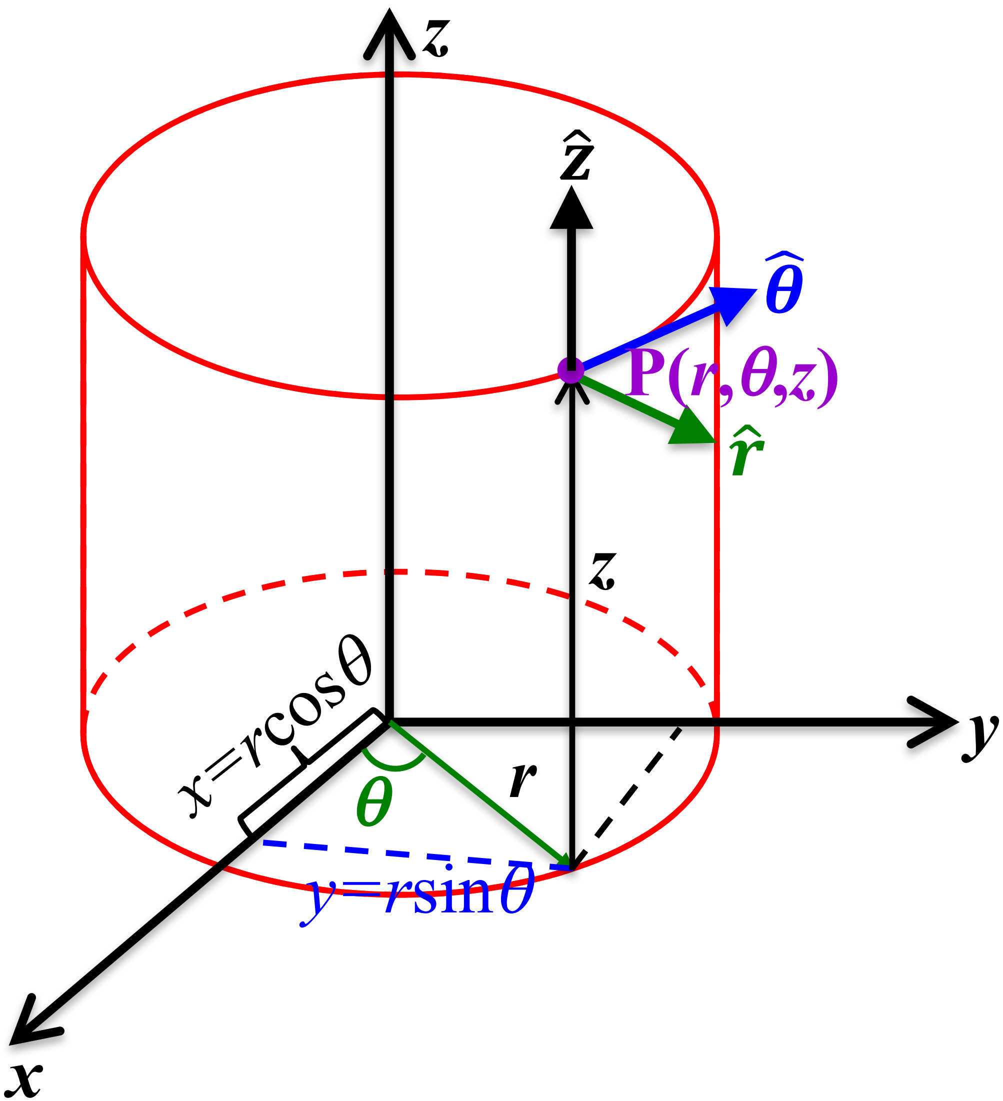




- In Cartesian coordinate system, one can define three unit vectors $\hat{x}$, $\hat{y}$ and $\hat{z}$ at every point.
- $\hat{x}$, $\hat{y}$ and $\hat{z}$ are unit vectors in the increasing direction of *x, y* and *z*.
- $\hat{x} \perp$ to the YZ plane, $\hat{y} \perp$ to the XZ plane and
  $\hat{z} \perp$ to the XY plane
- $\hat{x}$, $\hat{y}$ and $\hat{z}$ are orthogonal, and directed in the same direction at every points.

$$\hat{x} \times \hat{y} = \hat{z}\,;\ \hat{y} \times \hat{z} = \hat{x}\,;\ \hat{z} \times \hat{x} = \hat{y};$$

Cylindrical Coordinate System



How to locate a point 'P' in space ?

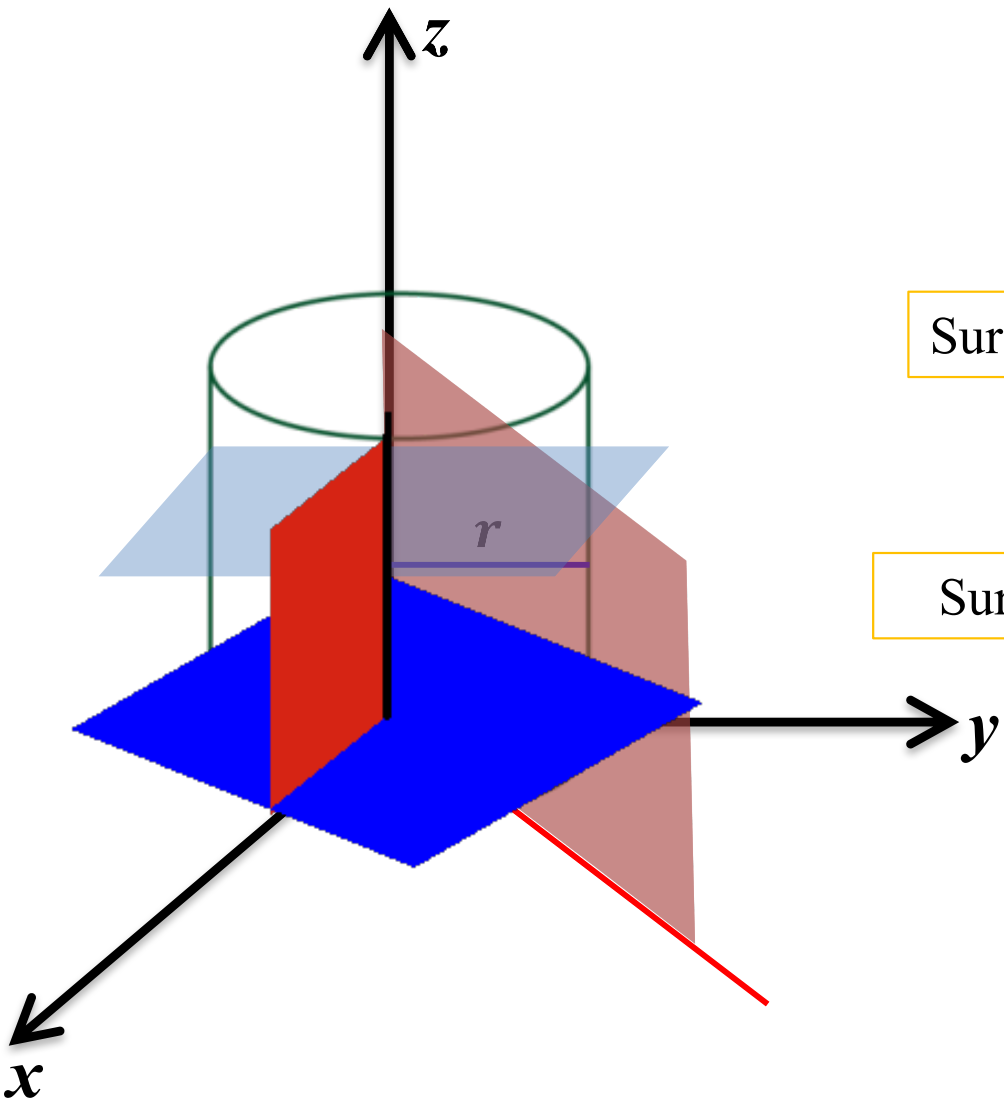
$(\boldsymbol{r}, \boldsymbol{\theta}, \boldsymbol{z})$ coordinates system is known as cylindrical coordinate system

$(\boldsymbol{r}, \boldsymbol{\theta})$ Coordinate of the foot of the point in XY plane

***z*-Height from the XY plane**

$\hat{\boldsymbol{r}}$, $\hat{\boldsymbol{\theta}}$, **and** $\hat{\boldsymbol{z}}$ are unit vectors along increasing direction of $\boldsymbol{r}$, $\boldsymbol{\theta}$, **and** $\boldsymbol{z}$.

Cylindrical Coordinate System



Note that point (r, θ, z) is at the intersection of three surfaces:

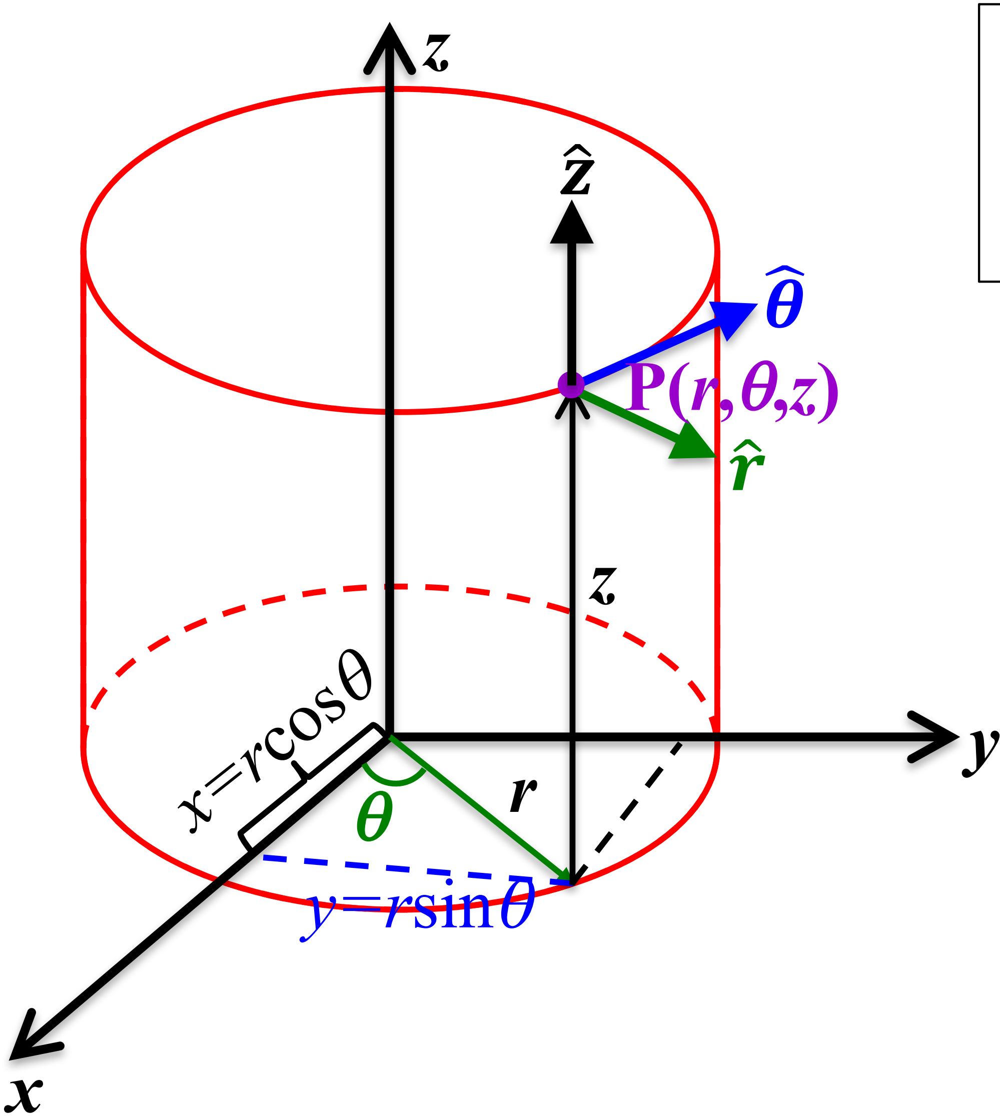
The surface with $\boldsymbol{r}$ =**Constant**; Cylinder

Surface with $\boldsymbol{\theta}$ = **Constant**; Plane containing Z-axis

Surface $\boldsymbol{z}$ = **Constant.** Plane Perpendicular Z-axis

The three orthogonal components, r (green), θ (red), and z (blue), each increasing at a constant rate. The point is at the intersection between the three colored surfaces.

Cylindrical Coordinate System



Transformation equation is very similar to polar coordinate with additional z-coordinate.

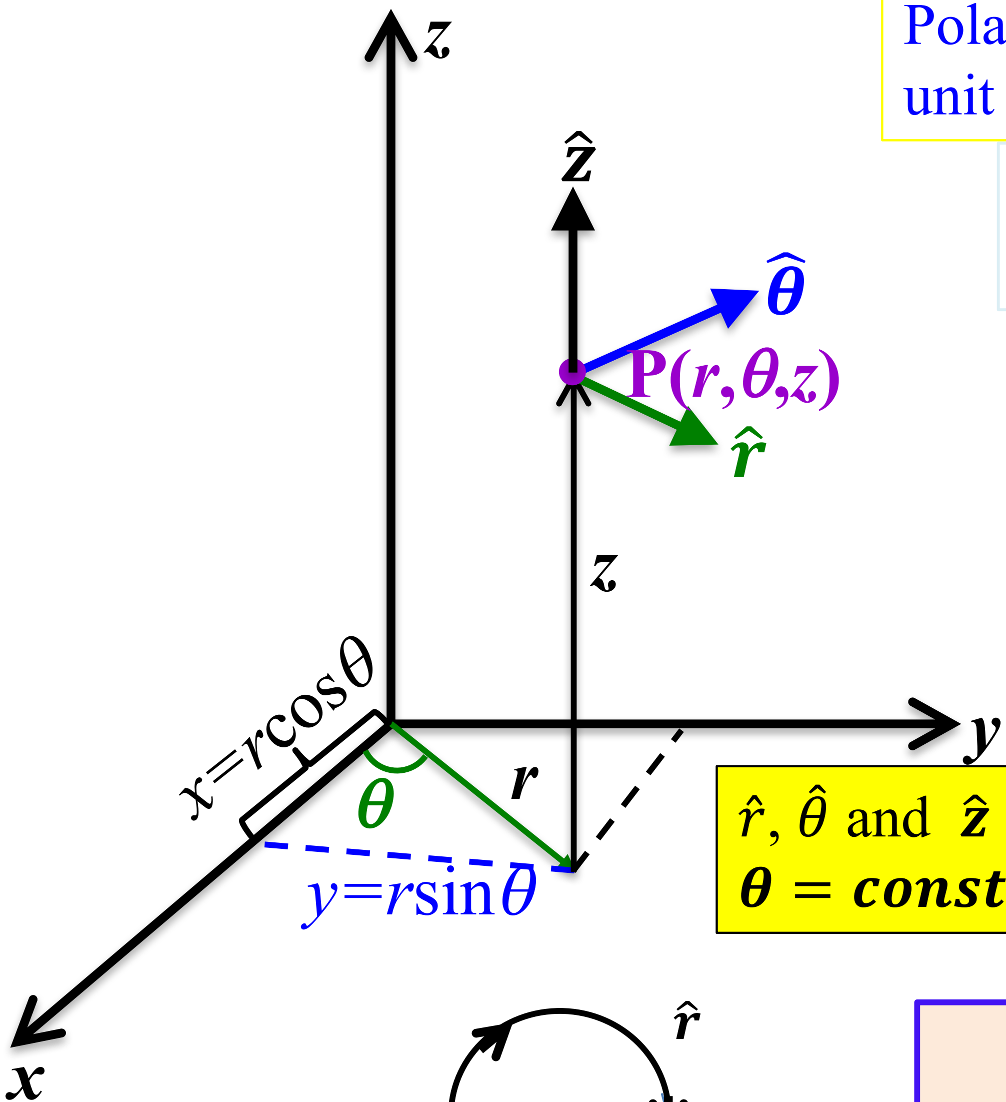
$$x = r\cos\theta$$
$$y = r\sin\theta$$
$$z = z$$

Reverse transformation

$$r = \left(x^2 + y^2\right)^{1/2}$$
$$\theta = \tan^{-1}\frac{y}{x}$$
$$z = z$$

Note: Instead of (r, θ), many books use notation (ρ, φ).

Cylindrical Coordinate System



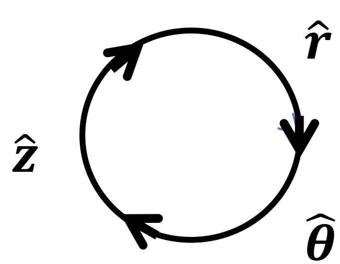
Polar coordinate unit vectors $(\hat{r}, \hat{\theta})$ + additional unit vector in the z −direction.

$\hat{\boldsymbol{r}}, \hat{\boldsymbol{\theta}}$ **and** $\hat{\mathbf{z}}$ are unit vectors along increasing direction of coordinates $\boldsymbol{r}, \boldsymbol{\theta}$ **and** $\boldsymbol{z}$.

$$\hat{\boldsymbol{r}} = \cos\boldsymbol{\theta}\ \hat{\boldsymbol{x}} + \sin\boldsymbol{\theta}\ \hat{\boldsymbol{y}}$$
$$\hat{\boldsymbol{\theta}} = -\sin\boldsymbol{\theta}\ \hat{\boldsymbol{x}} + \cos\boldsymbol{\theta}\ \hat{\boldsymbol{y}}$$
$$\hat{\mathbf{z}} = \hat{\mathbf{z}}$$

$\hat{r}$, $\hat{\theta}$, and $\hat{\mathbf{z}}$ are **orthogonal**, the directions of $\hat{r}$, $\hat{\theta}$, depend on location but $\hat{\mathbf{z}}$ is fixed

$\hat{r}$, $\hat{\theta}$ and $\hat{\mathbf{z}}$ are **perpendicular** to surfaces $\boldsymbol{r} = \boldsymbol{constant}$; $\boldsymbol{\theta} = \boldsymbol{constant}$ and $\boldsymbol{z} = \boldsymbol{constant}$, respectively.



$$\hat{\boldsymbol{r}} \times \hat{\boldsymbol{\theta}} = \hat{\mathbf{z}};\ \hat{\boldsymbol{\theta}} \times \hat{\mathbf{z}} = \hat{\boldsymbol{r}};\ \hat{\mathbf{z}} \times \hat{\boldsymbol{r}} = \hat{\boldsymbol{\theta}}$$

Position, Velocity, Acceleration, Newton's law in cylindrical coordinate system

Vector components are very similar to polar coordinate + z component

Position vector → $\vec{r} = r\hat{r} + \boldsymbol{z}\hat{\mathbf{z}}$

Velocity → $\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + \dot{\boldsymbol{z}}\hat{\mathbf{z}}$

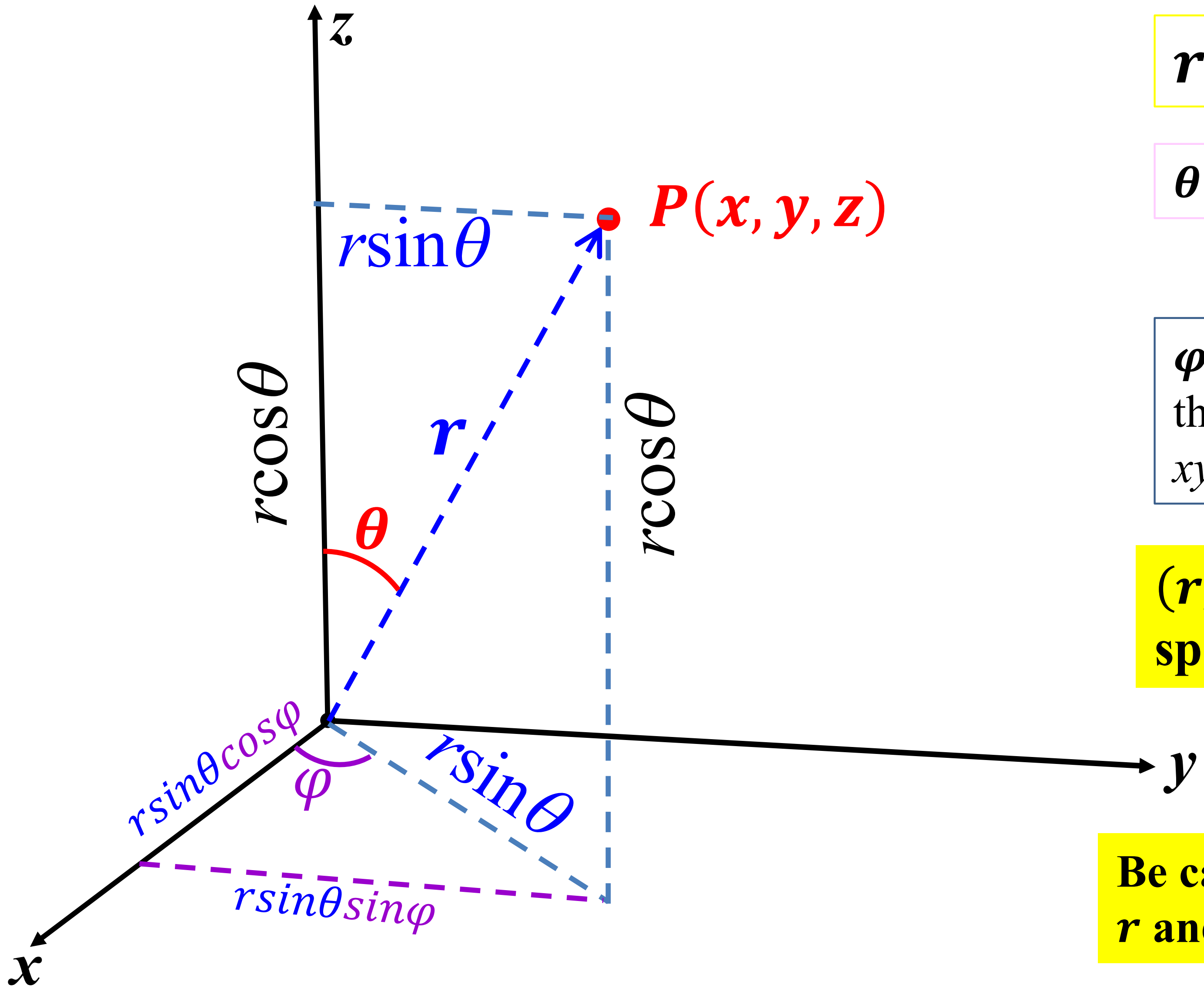
$$\frac{d\hat{\mathbf{z}}}{dt} = 0$$

Acceleration → $\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta} + \ddot{\boldsymbol{z}}\hat{\mathbf{z}}$

Newton's law →
$$\vec{F} = F_r\hat{r} + F_\theta\hat{\theta} + \boldsymbol{F_z}\hat{\mathbf{z}}$$
$$= m\left[\left(\ddot{r} - r\dot{\theta}^2\right)\hat{r} + \left(2\dot{r}\dot{\theta} + r\ddot{\theta}\right)\hat{\theta} + \ddot{\boldsymbol{z}}\hat{\mathbf{z}}\right]$$

Spherical polar coordinate

Spherical polar coordinate system



$\boldsymbol{r}$ → Radial distance from origin

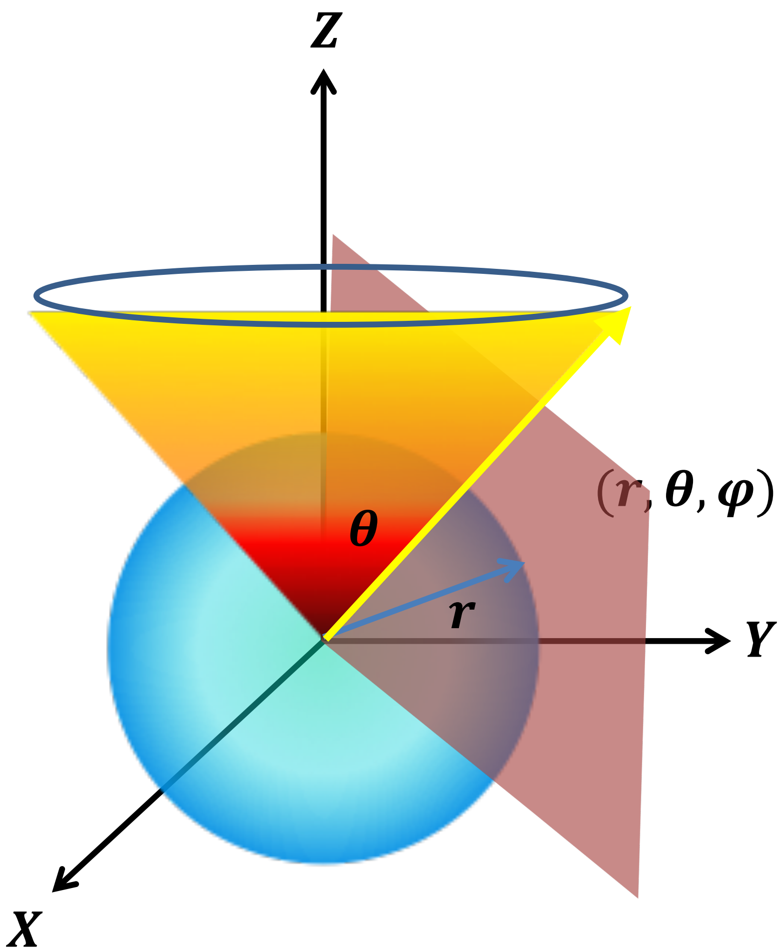
$\boldsymbol{\theta}$ → Angle with z-axis.

$\boldsymbol{\varphi}$ → Angle between x-axis and the projection of radial vector in xy plane

$(\boldsymbol{r}, \boldsymbol{\theta}, \boldsymbol{\varphi})$ is known as spherical polar coordinate

Be careful, notations are different. $\boldsymbol{r}$ and $\boldsymbol{\theta}$ are not planer coordinate.

Spherical polar coordinate system



Note that point (r, θ, φ) is at the intersection of three surfaces:

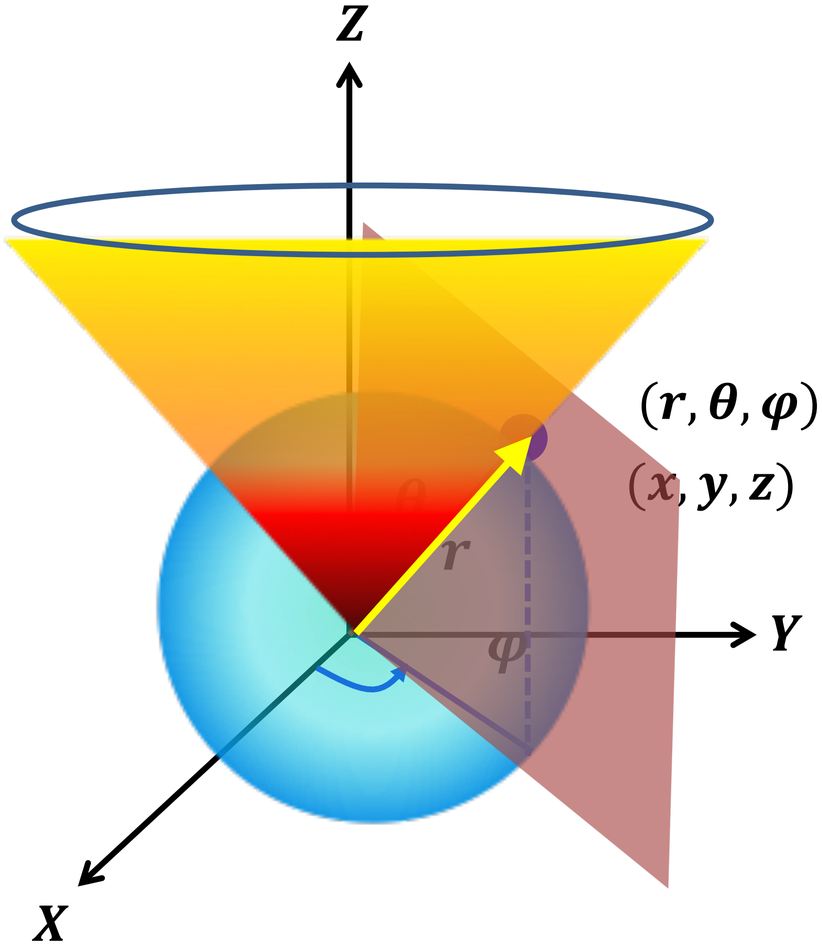
The surface with $\boldsymbol{r}$ =**Constant**; Sphere

Surface with $\boldsymbol{\theta}$ = **Constant**; Cone

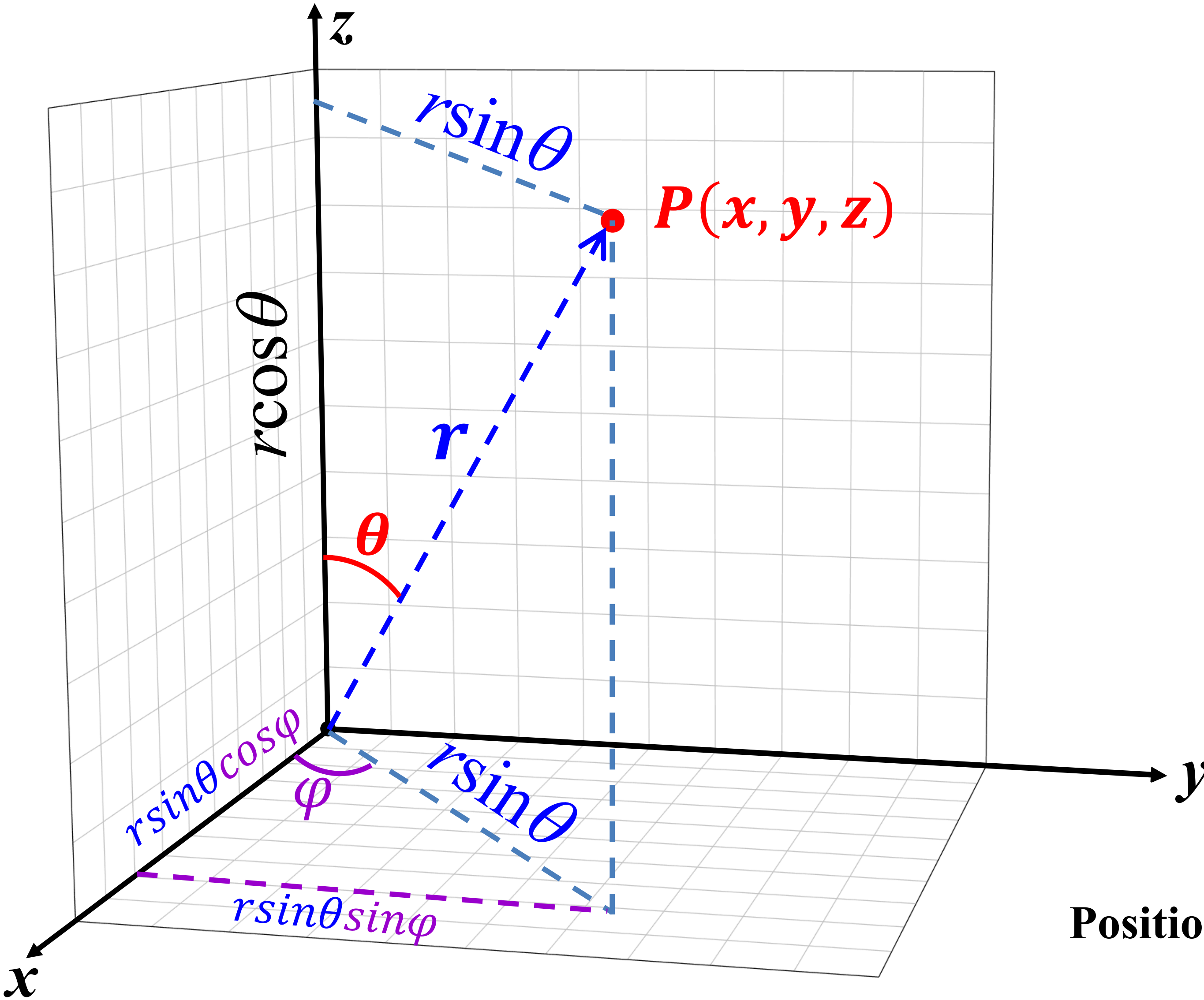
Surface $\boldsymbol{\varphi}$ = **Constant.** Plane containing Z-axis

Spherical polar coordinate system

So the point (r, θ, φ) is at the intersection of three surfaces:



Spherical polar coordinate system



Transformation relations

$$x = r\sin\theta\cos\varphi$$
$$y = r\sin\theta\sin\varphi$$
$$z = r\cos\theta$$

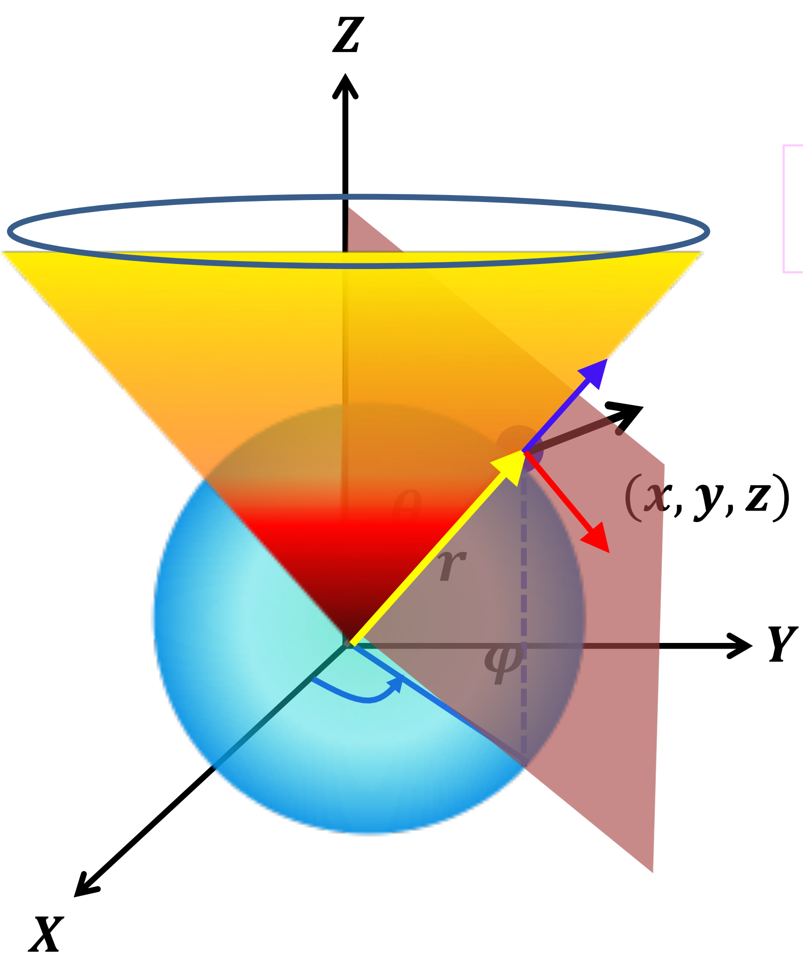
Hence

$$r = \left(x^2+y^2+z^2\right)^{1/2}$$
$$\theta = \tan^{-1}\left(\frac{\left(x^2+y^2\right)^{1/2}}{z}\right)$$
$$\varphi = \tan^{-1}\left(\frac{y}{x}\right)$$

Position vector in terms of x, y and z components

$$\vec{r} = r\sin\theta\cos\varphi\ \hat{x} + r\sin\theta\sin\varphi\ \hat{y} + r\cos\theta\ \hat{z}$$

Unit vectors: Spherical polar coordinate system



Position vector

$$\vec{r} = r\sin\theta\cos\varphi\,\hat{x} + r\sin\theta\sin\varphi\,\hat{y} + r\cos\theta\,\hat{z}$$

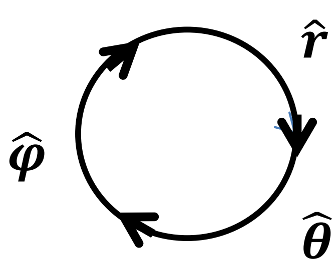
Unit vectors: $\hat{r} = \left(\frac{\partial\vec{r}/\partial r}{|\partial\vec{r}/\partial r|}\right)$, $\hat{\theta} = \left(\frac{\partial\vec{r}/\partial\theta}{|\partial\vec{r}/\partial\theta|}\right)$, $\hat{\varphi} = \left(\frac{\partial\vec{r}/\partial\varphi}{|\partial\vec{r}/\partial\varphi|}\right)$

$$\hat{r} = \sin\theta\cos\varphi\,\hat{x} + \sin\theta\sin\varphi\,\hat{y} + \cos\theta\,\hat{z}$$

$$\hat{\theta} = \cos\theta\cos\varphi\,\hat{x} + \cos\theta\sin\varphi\,\hat{y} - \sin\theta\,\hat{z}$$

$$\hat{\varphi} = -\sin\varphi\,\hat{x} + \cos\varphi\,\hat{y}$$

$\hat{r}$, $\hat{\theta}$, and $\hat{\varphi}$ are perpendicular to $r = constant$, $\theta = constant$, $\varphi = constant$.



We can also get the unit vectors

$$\hat{\boldsymbol{r}} = \hat{\boldsymbol{\theta}}\times\hat{\boldsymbol{\varphi}};\ \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\varphi}}\times\hat{\boldsymbol{r}};\ \hat{\boldsymbol{\varphi}} = \hat{\boldsymbol{r}}\times\hat{\boldsymbol{\theta}}$$

Unit vectors Spherical coordinate & Partial derivatives

Unit vectors in spherical polar coordinate are function of $\boldsymbol{\theta}$ and $\boldsymbol{\varphi}$ only.

$$\hat{r} = \sin\theta\cos\varphi\,\hat{x} + \sin\theta\sin\varphi\,\hat{y} + \cos\theta\,\hat{z}$$

$$\hat{\theta} = \cos\theta\cos\varphi\,\hat{x} + \cos\theta\sin\varphi\,\hat{y} - \sin\theta\,\hat{z}$$

$$\hat{\varphi} = -\sin\varphi\,\hat{x} + \cos\varphi\,\hat{y}$$

Thus,

$$\frac{\partial\hat{r}}{\partial r} = 0$$

$$\frac{\partial\hat{r}}{\partial\theta} = \frac{\partial}{\partial\theta}(\sin\theta\cos\varphi\,\hat{x} + \sin\theta\sin\varphi\,\hat{y} + \cos\theta\,\hat{z})$$

$$\frac{\partial\hat{r}}{\partial\theta} = \cos\theta\cos\varphi\,\hat{x} + \cos\theta\sin\varphi\,\hat{y} - \sin\theta\,\hat{z} = \hat{\theta}$$

$$\frac{\partial\hat{r}}{\partial\varphi} = \frac{\partial}{\partial\varphi}(\sin\theta\cos\varphi\,\hat{x} + \sin\theta\sin\varphi\,\hat{y} + \cos\theta\,\hat{z})$$

$$\frac{\partial\hat{r}}{\partial\varphi} = -\sin\theta\sin\varphi\,\hat{x} + \sin\theta\cos\varphi\,\hat{y} + 0 = \sin\theta\,\hat{\varphi}$$

$$\frac{\partial\hat{\theta}}{\partial r} = 0$$

$$\frac{\partial\hat{\theta}}{\partial\theta} = \frac{\partial}{\partial\theta}(\cos\theta\cos\varphi\,\hat{x} + \cos\theta\sin\varphi\,\hat{y} - \sin\theta\,\hat{z})$$

$$\frac{\partial\hat{\theta}}{\partial\theta} = (-\sin\theta\cos\varphi\,\hat{x} - \sin\theta\sin\varphi\,\hat{y} - \cos\theta\,\hat{z}) = -\hat{r}$$

Unit vectors Spherical coordinate & Partial derivatives

Unit vectors in spherical polar coordinate are function of θ and φ only.

$$\hat{r} = \sin\theta\cos\varphi\,\hat{x} + \sin\theta\sin\varphi\,\hat{y} + \cos\theta\,\hat{z}$$

$$\hat{\theta} = \cos\theta\cos\varphi\,\hat{x} + \cos\theta\sin\varphi\,\hat{y} - \sin\theta\,\hat{z}$$

$$\hat{\varphi} = -\sin\varphi\,\hat{x} + \cos\varphi\,\hat{y}$$

$$\frac{\partial\hat{\theta}}{\partial\varphi} = \frac{\partial}{\partial\varphi}(\cos\theta\cos\varphi\,\hat{x} + \cos\theta\sin\varphi\,\hat{y} - \hat{z}\sin\theta)$$

$$\frac{\partial\hat{\theta}}{\partial\varphi} = (-\cos\theta\sin\varphi\,\hat{x} + \cos\theta\cos\varphi\,\hat{y} - 0) = \cos\theta\,\hat{\varphi}$$

$$\boldsymbol{\frac{\partial\hat{\varphi}}{\partial r} = 0}$$

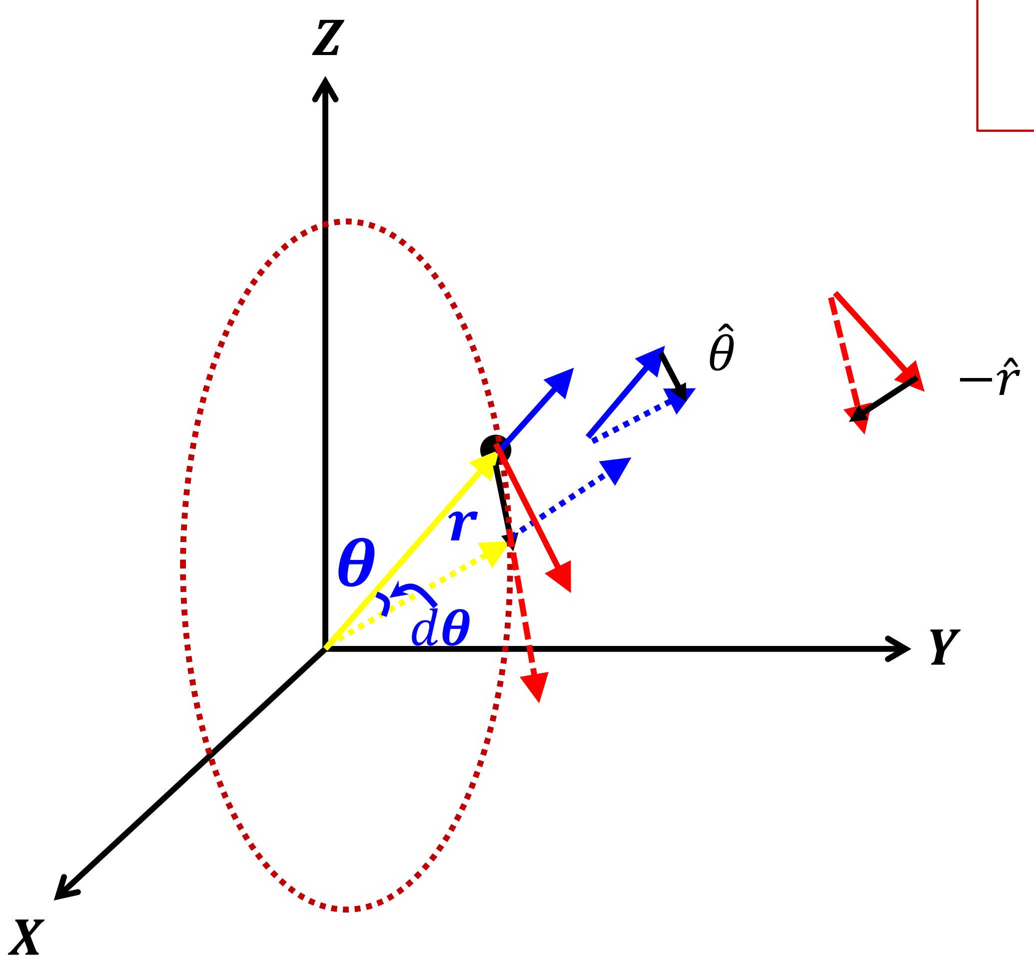
$$\frac{\partial\hat{\varphi}}{\partial\theta} = \frac{\partial}{\partial\theta}(-\sin\varphi\,\hat{x} + \cos\varphi\,\hat{y}) = 0$$

$$\frac{\partial\hat{\varphi}}{\partial\varphi} = \frac{\partial}{\partial\varphi}(-\sin\varphi\,\hat{x} + \cos\varphi\,\hat{y})$$

$$\frac{\partial\hat{\varphi}}{\partial\varphi} = -\cos\varphi\,\hat{x} - \sin\varphi\,\hat{y} = -(\cos\varphi\,\hat{x} + \sin\varphi\,\hat{y})$$

$$\frac{\partial\hat{\varphi}}{\partial\varphi} = -(\hat{r}\sin\theta + \hat{\theta}\cos\theta)$$

$$\boxed{\frac{\partial \hat{r}}{\partial \theta} = \hat{\theta}}$$



$$\frac{\partial \hat{\theta}}{\partial \theta} = -\hat{r}$$

Unit vectors Spherical coordinate & Partial derivatives

Unit vectors in spherical polar coordinate are function of θ and φ only.

$$\hat{r} = \sin\theta\cos\varphi\,\hat{x} + \sin\theta\sin\varphi\,\hat{y} + \cos\theta\,\hat{z}$$

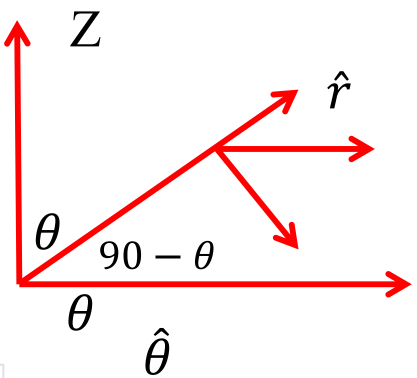
$$\hat{\theta} = \cos\theta\cos\varphi\,\hat{x} + \cos\theta\sin\varphi\,\hat{y} - \sin\theta\,\hat{z}$$

$$\hat{\varphi} = -\sin\varphi\,\hat{x} + \cos\varphi\,\hat{y}$$

$$\frac{\partial\hat{r}}{\partial r} = 0 \qquad \frac{\partial\hat{r}}{\partial\theta} = \hat{\theta} \qquad \frac{\partial\hat{r}}{\partial\varphi} = \sin\theta\,\hat{\varphi}$$

$$\frac{\partial\hat{\theta}}{\partial r} = 0 \qquad \frac{\partial\hat{\theta}}{\partial\theta} = -\hat{r} \qquad \frac{\partial\hat{\theta}}{\partial\varphi} = \cos\theta\,\hat{\varphi}$$

$$\frac{\partial\hat{\varphi}}{\partial r} = 0 \qquad \frac{\partial\hat{\varphi}}{\partial\theta} = 0 \qquad \frac{\partial\hat{\varphi}}{\partial\varphi} = -\left(\hat{r}\sin\theta + \hat{\theta}\cos\theta\right)$$



A line in XY plane at angle φ with x-axis

$$-\sin\varphi\,\hat{x} + \cos\varphi\,\hat{y} = \hat{\varphi}$$

$$\cos\varphi\,\hat{x} + \sin\varphi\,\hat{y} = \sin\theta\,\hat{r} + \cos\theta\,\hat{\theta}$$

$$\hat{x} = \sin\theta\cos\varphi\,\hat{r} + \cos\theta\cos\varphi\,\hat{\theta} - \sin\varphi\,\hat{\varphi}$$

$$\hat{y} = \sin\theta\sin\varphi\,\hat{r} + \cos\theta\sin\varphi\,\hat{\theta} + \cos\varphi\,\hat{\varphi}$$

$$\hat{z} = \cos\theta\,\hat{r} - \sin\theta\,\hat{\theta}$$

Velocity in Spherical coordinate

$$\hat{r} = \sin\theta\cos\varphi\,\hat{x} + \sin\theta\sin\varphi\,\hat{y} + \cos\theta\,\hat{z}$$

$$\hat{\theta} = \cos\theta\cos\varphi\,\hat{x} + \cos\theta\sin\varphi\,\hat{y} - \sin\theta\,\hat{z}$$

$$\hat{\varphi} = -\sin\varphi\,\hat{x} + \cos\varphi\,\hat{y}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(r\hat{r})$$

$$= \frac{dr}{dt}\hat{r} + r\frac{d\hat{r}}{dt}$$

$$\frac{\partial\hat{r}}{\partial\theta} = \hat{\theta} \qquad \frac{\partial\hat{r}}{\partial\varphi} = \sin\theta\,\hat{\varphi}$$

$$= \dot{r}\hat{r} + r\left(\frac{\partial\hat{r}}{\partial\theta}\frac{d\theta}{dt} + \frac{\partial\hat{r}}{\partial\varphi}\frac{d\varphi}{dt}\right) \Leftarrow \text{Chain rule}$$

$$\vec{v} = \dot{r}\hat{r} + r\left(\dot{\theta}\hat{\theta} + \sin\theta\,\dot{\varphi}\,\hat{\varphi}\right)$$

$$\boldsymbol{\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\sin\theta\,\dot{\varphi}\,\hat{\varphi}}$$

You need to try this out?

Can you obtain Acceleration??

$$\vec{a} = \left(\ddot{r} - r\dot{\theta}^2 - r\dot{\varphi}^2\sin^2\theta\right)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta} - r\dot{\varphi}^2\sin\theta\cos\theta)\hat{\theta} + \left(r\ddot{\varphi}\sin\theta + 2r\dot{\theta}\dot{\varphi}\cos\theta + 2\dot{r}\dot{\varphi}\sin\theta\right)\hat{\varphi}$$

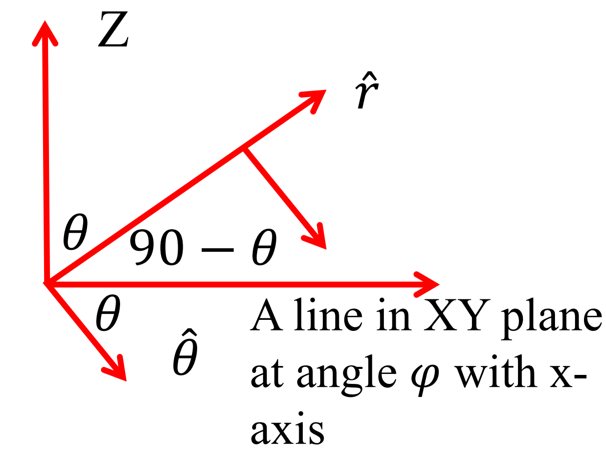
Velocity in Spherical coordinate

$$\vec{a} = \left(\ddot{r} - r\dot{\theta}^2 - r\dot{\varphi}^2\sin^2\theta\right)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta} - r\dot{\varphi}^2 \sin\theta\cos\theta)\hat{\theta} + \left(r\ddot{\varphi}\sin\theta + 2r\dot{\theta}\dot{\varphi}\cos\theta + 2\dot{r}\dot{\varphi}\sin\theta\right)\hat{\varphi}$$

$$a_r = \left(\ddot{r} - r\dot{\theta}^2 - r\sin\theta\dot{\varphi}^2\sin\theta\right)$$

Centripetal part due to angular motion in a plane containing Z- axis

Centripetal part due to angular motion in xy plane



$$a_\theta = (r\ddot{\theta} + 2\dot{r}\dot{\theta} - r\sin\theta\dot{\varphi}^2\cos\theta)$$

Coriolis part due to angular motion in a plane containing Z- axis

Centripetal part due to angular motion in xy plane

$$a_\varphi = \left(r\ddot{\varphi}\sin\theta + 2r\dot{\theta}cos\theta\dot{\varphi} + 2\dot{r}sin\theta\dot{\varphi}\right)$$

Coriolis part due to angular motion in xy plane